

ANUSHA AGRAWAL

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Education

University of Massachusetts, Amherst | Master of Science in Computer Science GPA: 3.9/4 | Sept 2024 – May 2026

Savitribai Phule Pune University | B.Tech Electronics

GPA: 3.7/4 | Sept 2019 – May 2023

Experience

Microsoft

Feb 2026 – Present

AI Research Extern

Remote

- (under review: **ICLR 2027**) Developed entropy-based **adaptive LLM** inference for LLaMA-3.1-8B using ~1B Pile training tokens, reducing autoregressive generation steps by **2.3x** and improving wall-clock speed by **2.1x**
- Built a **PyTorch** harness running latency, throughput, and perplexity checks per LLM checkpoint, catching regressions before full evaluation runs
- Implemented **GPU-batched** model analysis tooling for token-level entropy scoring, replacing sequential prefix scoring with full-sequence inference, reducing evaluation runtime from **90s to 3.3s/sample**

UMass BioNLP Research Lab

Jul 2025 – Dec 2025

Graduate Student Researcher

Amherst, MA

- Engineered an **LLM-based medical calculation** pipeline to extract calculator inputs from patient notes, flagging missing evidence, and executing complete cases with deterministic Python tools
- Re-evaluated LLM-based clinical calculators with **stepwise workflows**, finding accuracy inflated by **20** points; improved reliability via **RAG** formula selection and scaled evaluation with an **LLM-as-judge** at **95%+** clinician agreement
- Curated **80+** source-backed, **multi-calculator** tasks and normalized shared clinical parameters into a canonical schema for retrieval, structured tool use, missing-information detection, and chained reasoning

Oracle

Jul 2023 – Jul 2024

Software Engineer

Pune, India

- Traced **12%** P95 latency regression in **Java** (Spring Boot) billing services handling **10M+** daily records to N+1 query patterns in REST API layer, fixing it by batching fetches
- Indexed the reconciliation table after **SQL** query plans exposed full table scans behind month-end batch delays, cutting peak-cycle latency **20%** across **300M+** monthly transactions
- Partnered with enterprise clients to translate CRM and billing requirements into technical designs and **JavaScript** dashboards across **5M+** monthly billing events, reducing discrepancy resolution time by **30%**
- Debugged recurring 502 spike in **Kubernetes**/Istio-managed microservices, ruling out network-policy issue before isolating sidecar misconfiguration and fixing readiness probe, stabilizing rolling deploys
- Leveraged Copilot to draft regression tests, catching assertions that missed real bugs and adding edge cases like null and duplicate inputs, reducing release defects by **26%**

Projects

AI-Assisted Code Collaboration Platform | Go, React, TypeScript, WebSockets, Redis, Docker, AWS, LLM Jan 2025

- Built real-time collaborative coding platform in **Go** and **React** with CRDT-based sync over WebSockets and Redis-backed session state, reaching sub-**100ms** latency.
- Added LLM-powered explanations and test generation, validating against shared repo, cutting debugging time **30%**
- Containerized services with **Docker** and Kubernetes on **AWS** with health checks and reconnect handling, supporting 10K+ simulated editing sessions

Low-Latency Distributed Storage and Metadata Service | Rust, C++, Linux, Cassandra, Hadoop

Dec 2025

- Built **distributed storage** across **20+** simulated nodes using consistent hashing, partitioned metadata, replication, and node-failure recovery, improving availability by **41%**
- Indexed metadata lookups by partition, reducing latency from **185ms to 111ms** and increasing throughput from **2.4K to 4.8K ops/sec**

Technical Skills

Programming: Python, Java, C++, Go, Rust, JavaScript, SQL, Bash

Full-Stack: React, Spring Boot, JPA, REST APIs, gRPC, Microservices, WebSockets, API Design, Service Integration

Systems, Cloud & Data: Distributed Systems, Multithreading, Linux, AWS, Azure, Docker, Kubernetes, Istio, CI/CD, SLURM, Kafka, PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, Redis, Hadoop, PySpark

AI/ML & Developer Tools: PyTorch, Transformers, vLLM, Hugging Face, LLM APIs, RAG, FAISS, SBERT, LLM-as-Judge, Model Evaluation, Benchmarking, MLflow, Git